MILITARY SPECIFICATION

REFRIGERATORS AND RELATED EQUIPMENT, PACKAGING AND PACKING OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers optional detailed requirements or performance requirements for preservation, packaging, packing, and marking of mechanically cooled refrigerators, freezers, and related equipment with storage volume up to 100 cubic feet.

1.2 Classification.

1.2.1 Options (see 3.5.1, 3.6.1 and 3.6.2).

Option I - Detailed requirements.
Option II - Performance requirements.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
NN-P-530	-	Plywood, Flat Panel
QQ-S-781	-	Strapping, Steel, and Seals
TT-P-664	-	Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting
UU-P-134	-	Paper, Wrapping, Wet-Waxed
UU-T-81	-	Tags, Shipping and Stock

FSC 4110

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PPP-B-591 - Boxes, Shipping, Wood-Cleated
PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-640 - Box, Fiberboard, Corrugated, Triple-Wall
PPP-B-1055 - Barrier Material, Waterproofed, Flexible
PPP-C-650 - Crates, Wood, Open and Covered
PPP-C-843 - Cushioning Material, Cellulosic
PPP-P-291 - Paperboard, Wrapping and Cushioning
PPP-P-600 - Porcelain Enamel Products and Household Appliances, Electrical and Mechanical - Requirements for Packaging and Packing
PPP-S-760 - Strapping, Nonmetallic, (and Connectors)
PPP-T-60 - Tape: Packaging, Waterproof
PPP-T-97 - Tape; Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116 - Preservation-Packaging, Methods Of
MIL-C-11133 - Crate, Wood, Open, Wirebound

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-731 - Quality of Wood Members for Containers and Pallets

DRAWINGS

US ARMY NATICK DEVELOPMENT CENTER

5-13-1153 - Refrigerators and Related Equipment, Packaging and Packing of: Container, Cleated-Plywood, and Container Cleated Fiberboard, for Levels A and B Shipments
5-13-1172 - Refrigerators and Related Equipment, Packaging and Packing of: Method of Securing Units Within Exterior Shipping Containers

(Miniature reproductions of Drawings 5-13-1153 and 5-13-1172 identified as figures 1 and 2 respectively, are attached for information purposes only.)

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring agency, or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N. W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Department of Commerce

PS 1 - Softwood Plywood Construction and Industrial
PS 51 - Hardwood and Decorative Plywood

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Materials. Materials used for cleaning, preservation, packaging, packing, and marking shall conform to the applicable specifications as specified herein.

3.2 Workmanship. Preservation, packaging, packing, and marking shall conform to the quality and grade of product established by this specification.

3.3 Disassembly and matchmarking.

3.3.1 Disassembly. All detachable parts and accessories, such as trays, shelves, food jars, and other parts liable to be damaged or cause damage during handling and shipment, shall be removed from the unit and packaged as specified in 3.5.1.1.11. Bolts, nuts, screws, washers, pins, and other connectors removed during disassembly shall be placed in a cotton cloth drawstring bag and tied

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to the unit or disassembled part, or packaged in the same fiberboard box as the accessories and parts. Alternatively, detached parts and accessories for units packed as specified in option II, may be secured in their normal position in a manner to prevent damage.

3.3.2 Matchmarking. When, in the opinion of the manufacturer, detached parts and accessories may cause difficulty in reassembly, they shall be matchmarked. Matchmarking information shall be on shipping tags conforming to type A of UU-T-81, attached to mating parts.

3.4 Draining. All water-containing parts, such as automatic ice cube makers, water cooling systems, and water-cooled condensers, shall be completely drained to prevent damage from freezing.

3.5 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

3.5.1 Level A. Units with a combined gross weight (complete unit and shipping container) not exceeding 500 pounds shall be preserved and packaged either option I or II at the discretion of the supplier. Units with a combined gross weight exceeding 500 pounds shall be preserved and packaged option I (see 1.2.1 and 6.1).

3.5.1.1 Option I - detailed requirements. Cleaning, application and type of preservative, and method of preservation shall be in accordance with MIL-P-116 as specified herein.

3.5.1.1.1 Cleaning. All exposed, uncoated, ferrous metal surfaces of the unit, detached parts, and accessories shall be thoroughly cleaned in accordance with process C-1 and thoroughly dried.

3.5.1.1.2 Application of preservative. All exposed, uncoated, ferrous metal surfaces, except corrosion-resisting steel, within the food storage or freezer compartments, shall be coated with type P-14 preservative. All other exposed uncoated, ferrous metal surfaces, except corrosion-resisting steel, shall be coated with type P-1 preservative. Pulleys and shafts shall be coated as specified in 3.5.1.1.7.

3.5.1.1.3 Protection of preservative. Surfaces coated with type P-14 preservative from which preservative might be removed by contact with packaging and packing materials, shall be covered, wrapped, or bagged with greaseproof barrier material in accordance with method I.

3.5.1.1.4 Sealing. All exposed ends of wires, plug openings, and sockets shall be sealed with tape conforming to type III, class 1 of PPP-T-60.

3.5.1.1.5 Electrical components. Detached electrical components, such as motors, generators, control boxes, and temperature and pressure control switches shall be preserved by method II, with selection of the submethod at the option

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of the supplier. Attached electrical components not inclosed within the storage compartment of the refrigerator shall have all openings into their interiors, such as inlets, outlets, vents, valves, and louvers sealed with tape as specified in 3.5.1.1.4.

3.5.1.1.6 Operating instruction. Operating instructions and service manuals shall be individually placed within a commercial paper envelope and secured within the interior of the unit with pressure sensitive tape or other affixing means.

3.5.1.1.7 Servicing of refrigerating unit. The refrigerating system shall be serviced for shipment according to the manufacturer's recommended procedure and as specified herein. Drive belts shall be removed, and the exposed, uncoated, ferrous metal surfaces of pulleys and shafts cleaned in accordance with process C-1 and coated with primer conforming to TT-P-664. When primer is thoroughly dry, drive belts shall be remounted in place with tension on the belts relaxed. All floating mechanisms and tubing not rigidly fixed, such as vibration loops, free coils, capillary tube loops and coils, shall be secured in a fixed position to minimize vibration and subsequent damage during handling and shipment. Floating mechanism is defined as any part that is inherently flexible or movable, or otherwise mounted which might vibrate during shipment.

* 3.5.1.1.8 Doors, drawers, and bins. Interior doors, drawers, and bins not covered in 3.3.1 shall be secured in the closed position in a manner to prevent movement or damage while in transit. Exterior doors, drawers, and bins of the unit shall be secured by one of the following methods:

(a) Locking. When means of locking is provided, the doors, drawers, and bins shall be closed, latched, and locked in a secure position. When locking is by means of a padlock, the doors and bins shall be closed and locked by means of a bolt and nut in a manner to prevent them from opening. When doors, drawers, and bins are provided with built in locks they shall be closed, locked, and the keys placed in a cotton cloth drawstring bag and the bag secured to one of the handles. The bagged keys and handle shall then be wrapped with tape specified in 3.5.1.1.4.

(b) Strapping. When means of locking doors, drawers, and bins are not provided, they shall be closed and secured with steel strapping conforming to class 1, type I of QQ-S-781 or nonmetallic strapping conforming to type II or III of PPP-S-760. Steel strapping shall be minimum 1/4 by 0.015 inch on household type refrigerators and minimum 3/8 by 0.020 inch on all other type units positioned above or near the latches. The use and positioning shall be the same for nonmetallic strapping. A continuous strip of cushioning material minimum 1-1/2 inches wide and 1/2 inch thick, conforming to type II, class B of PPP-C-843, shall be positioned under the steel strapping. The nonmetallic strapping shall be

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minimum 3/8 inch by 0.015 inch on all other type units positioned above or near the latches. The strapping shall be centered on the cushioning material and shall completely encircle the unit in a horizontal position to prevent contact of the steel strapping with the finished surfaces of the unit. The clips or buckles on nonmetallic strapping shall not bear on the finished surfaces of the front, sides, or top of the unit. Tensioning of strapping shall not mar or distort the surface of the unit.

(c) Taping. Alternative to strapping, the doors, drawers, or bins shall be secured in a closed position with 1 inch minimum width tape conforming to type III of PPP-T-97 extending around the entire unit and overlapping itself not less than six inches, or by not less than four strips of tape, each a minimum of 18 inches in length, applied in an "X" position to the latch or unhinged side of the door and side. The tape shall show no staining when tested in accordance with 4.4.1.

3.5.1.1.9 Legs. Detachable legs shall be removed from the unit and, together with attaching hardware, shall be cushioned and packaged as specified in 3.5.1.1.11. Leveling pads on detachable legs shall be completely screwed into the legs. Permanently attached legs shall have the leveling pads removed (unless pads are permanently assembled) and packaged with accessories and parts.

3.5.1.1.10 Line cord. The nondetachable line cord shall be looped and securely tied in place at the relay bracket, or similar location, or to the back of the unit. A voltage and frequency notice shall be secured to the cord, except when such information appears on the manufacturer's nameplate.

* 3.5.1.1.11 Accessories and parts. Glass shelves, trays, food jars, covers, and similar fragile items, metal or wire shelves, accessories, and parts shall be individually wrapped with cushioning material conforming to PPP-C-843 or PPP-P-291 and packaged in a fiberboard box or boxes conforming to grade V3c, V3s or V4s, style optional, of PPP-B-636. Each box shall be closed in accordance with method III as specified in the appendix of PPP-B-636. When total gross weight of the packaged accessories and parts is 65 pounds or less, the fiberboard box or boxes shall be secured or blocked and braced within the unit in a manner to prevent movement during handling and shipment. When adequate means are not available to secure the fiberboard box within the unit, or when total gross weight exceeds 65 pounds, the accessories and parts shall be packed as specified in 3.6.4.

3.5.1.1.12 Tags. A tag or tags conforming to type A of UU-T-81 shall be attached in a conspicuous place on each unit and give detailed instructions for deprocessing or placing the unit into service.

3.5.1.2 Option II - performance requirements.

3.5.1.2.1 Preservation and packaging. Preservation and packaging shall be as specified in 3.5.1.1 through 3.5.1.1.7, 3.5.1.1.9, 3.5.1.1.10, and 3.5.1.1.12.

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3.5.1.2.2 Doors, drawers, and bins. Doors, drawers, and bins shall be packaged as specified in 3.5.1.1.8 or in a manner to prevent movement or damage to the component or unit during shipment or handling. As a minimum, there shall be no evidence of damage to these components when the packed unit is tested for the applicable level of pack in PPP-P-600.

3.5.1.2.3 Accessories and parts. Accessories and parts shall be packaged as specified in 3.5.1.1.11, or they may be secured in their normal position or relocated within the unit and secured to prevent damage to the part or unit. Accessories and parts, secured in place or relocated within the unit, shall not be damaged or cause damage to the unit when tested as specified for the applicable level of pack of PPP-P-600.

* 3.5.2 Level C. The units shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

3.6 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

3.6.1 Level A. Units with a combined gross weight (complete unit and shipping container) not exceeding 500 pounds shall be packed either option I or II at the discretion of the supplier. Units with a combined gross weight exceeding 500 pounds shall be packed option I (see 1.2.1 and 6.1).

* 3.6.1.1 Option I - detailed requirements. Each unit, together with accessories and parts, preserved and packaged as specified in 3.5.1.1 or 3.5.2, shall be packed in a shipping container as specified in table I.

* 3.6.1.1.1 Boxes, fiberboard, wood cleated - maximum weight of contents 200 pounds. Wood-cleated, fiberboard boxes shall conform to class II, style A or B of PPP-B-591, except as shown on Drawing 5-13-1153. Each shipping container shall be strapped as shown on Drawing 5-13-1153.

* 3.6.1.1.2 Boxes, wood, cleated-plywood - maximum weight of contents 800 pounds. Wood-cleated plywood boxes shall conform to overseas type, style A or B, Grade A or B, type 2 load of PPP-B-601, except as shown on Drawing 5-13-1153. Each shipping container shall be strapped as shown on Drawing 5-13-1153.

* 3.6.1.1.3 Boxes, wood, nailed and lock corner - maximum weight of contents 800 pounds. Wood boxes shall conform to class 2, style 7, grade A or B, type 3 load of PPP-B-621, except that reinforcing members shall be of sufficient width to insure complete contact with fiberboard edge pads when refrigerators are secured method A or B. The top, sides, and ends of the container shall be lined with waterproof barrier material conforming to PPP-B-1055, with all joints having a 4-inch minimum single lap.

* 3.6.1.1.4 Crates, wood, lumber and plywood sheathed, nailed and bolted, weight of contents exceeding 500 pounds. Sheathed nailed wood crates shall conform to type I or II, class 1 or 2, style a of MIL-C-104, except for weight of contents up to and including 1000 pounds the following shall apply:

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(a) All structural members of the end panels shall be minimum 1- by 4-inch (nominal) lumber.

(b) All structural members of the side panels, except the vertical end struts, shall be nominal 1- by 4-inch lumber. The vertical end struts shall be nominal 2- by 4-inch lumber.

(c) A continuous sheet of minimum 1/2-inch thick plywood may be substituted for the 1/4-inch thick plywood, roofing felt, and wood sheathing on the top of narrow top plywood sheathed crates. The plywood shall conform to NN-P-530, either PS 51, type I, grade 3-4 or PS 1, type interior, grade C-D with exterior glue.

(d) The bolts used to prevent splitting of the 2- by 4-inch skids shall be minimum 1/4-inch diameter or maximum 3/8-inch diameter.

3.6.1.2 Option II - performance requirements. Units, preserved and packaged as specified in 3.5.1.2 or 3.5.2, shall be packed in accordance with the level A requirements of PPP-P-600. The packed unit shall meet the level A requirements of PPP-P-600.

3.6.2 Level B. Units with a combined gross weight (complete unit and shipping container) not exceeding 500 pounds shall be packed either option I or II at the discretion of the supplier. Units with a combined gross weight exceeding 500 pounds shall be packed option I (see 1.2.1 and 6.1).

3.6.2.1 Option I - detailed requirements. Each unit, together with accessories and parts, preserved and packaged as specified in 3.5.1.1 or 3.5.2, shall be packed in a shipping container as specified in table I. The contents of each open type crate shall be protected from the elements by a full length shroud fabricated from minimum 4 mil thick polyethylene conforming to L-P-378 and the seams shall be heat sealed. The shroud shall hang free, when possible, and shall extend to the base of the crate. All sharp points of contact between the item, the blocking and bracing, and the shroud shall be cushioned to prevent rupture or chafing of the shroud. All open type crates shall be assembled, closed, and strapped in accordance with the appendix of the applicable crate specifications.

3.6.2.1.1 Boxes, fiberboard. The fiberboard shipping containers shall be snug-fitting conforming to style RSC, type CF or SF, class domestic of PPP-B-636. The contents shall be immobilized by the use of die-cut or built-up fiberboard pads or cells. Each fiberboard shipping container shall be securely closed in accordance with method II as specified in the appendix of the container specification. When specified (see 6.2), the fiberboard shipping containers shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification.

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3.6.2.1.2 Boxes, fiberboard, corrugated, triple-wall-maximum weight of contents 275 pounds. Triple-wall, corrugated, fiberboard boxes shall conform to class 1, grade A, style E of PPP-B-640. The contents shall be immobilized by the use of die-cut or built-up fiberboard pads or cells made of material specified in 3.6.2.1.1. Each fiberboard shipping container shall be securely closed in accordance with the appendix of the container specification. When specified (see 6.2), the triple-wall corrugated shipping containers shall be fabricated of class 2; weather-resistant material.

3.6.2.1.3 Boxes, fiberboard, wood-cleated-maximum weight of contents 400 pounds. Wood-cleated, fiberboard, shipping containers shall conform to class I of PPP-B-591 and as shown on Drawing 5-13-1153. When specified (see 6.2), class II weather-resistant material shall be used in the fabrication of the shipping container.

3.6.2.1.4 Boxes, wood, cleated-plywood-maximum weight of contents 700 pounds. Wood-cleated, plywood, shipping containers shall conform to domestic type, grade B of PPP-B-601 and as shown on Drawing 5-13-1153.

3.6.2.1.5 Crate, wood, open, wirebound-maximum weight of contents 1,000 pounds Wirebound crates shall conform to type II, class 1 or 2, style 1 or 2, grade B; or type III, class 1, 2, or 3, style 1 or 2, grade B of MIL-C-11133, as applicable.

3.6.2.1.6 Crates, wood, open and covered-maximum weight of contents 2,500 pounds. Open wood crates shall conform to style A, type I, II, IV, or V, class 1 of PPP-C-650, as applicable.

3.6.2.1.7 Crates, wood, lumber and plywood sheathed, nailed and bolted-weight of contents exceeding 500 pounds. Sheathed, nailed, wood crates shall be as specified in 3.6.1.1.4, except that the waterproof barrier material shall not be required.

3.6.2.2 Option II - performance requirements. Units, preserved and packaged as specified in 3.5.1.2 or 3.5.2, shall be packed in accordance with the level B requirements of PPP-P-600. The packed unit shall meet the level B performance requirements of PPP-P-600.

* 3.6.3 Level C. Units, preserved and packaged as specified in 3.5, shall be secured and packed in shipping containers in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rates for such supplies. Containers and packing shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

3.6.4 Securing of units in the shipping container-option I and II. Each complete unit shall be secured within one shipping container by the methods as indicated in table I for option I for the weight limitations specified, except for level C shipments. Each complete unit shall be secured in a manner to

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prevent movement or damage for option II. A minimum of 1 inch clearance shall be provided between the top, sides, and ends of the unit and the adjacent inside surfaces of the shipping container. The fiberboard box or boxes containing accessories and parts not secured within the unit shall be secured in an unused portion of the same shipping container in a manner to prevent movement or damage to the accessories, parts, or unit during transit or storage.

TABLE I. Methods of securing units in shipping containers and maximum weight of contents for option I-Level A and B

Container	Securing method	Weight of contents (pounds)
Boxes, shipping, fiberboard, PPP-B-636 1/	See 3.6.2.1.1	Weight of content shall not exceed the weight limitation of the specification
Boxes, fiberboard, corrugated 1/ triple-wall, PPP-B-640	See 3.6.2.1.2	275 Maximum
Boxes, shipping, fiberboard, wood-cleated PPP-B-591 and Drawing No. 5-13-1153	B 3/ A or C 3/ or A and C 3/	300 Maximum 2/ 400 Maximum 2/
Boxes, wood, cleated-plywood, PPP-B-601 and Drawing No. 5-13-1153	B 3/ A or C 3/ or A and C 3/	300 Maximum 700 Maximum
Boxes, wood, nailed and locked corner - PPP-B-621	B or C	800 Maximum
Crates, wood; lumber and plywood sheathed, nailed and bolted, MIL-C-104	A or C 3/ or A and C 3/	Exceeding 500
Crate, wood, open, wirebound 1/ MIL-C-11133	B A or C or A and C	300 Maximum 1,000 Maximum
Crates, wood, open and 1/ covered, PPP-C-650	B A or C or A and C	300 Maximum 2,500 Maximum

1/ For level B shipments only.

2/ 200 maximum for level A shipment.

3/ Cushioning material shall be water-resistant for level A shipment.

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3.6.4.1 Method A (bolting). The unit shall be bolted to the base and one panel of the container and shall be cushioned over the full length of the two front-side edges of the unit. A minimum 1-inch clearance shall be provided between the outermost projections of the front, sides, and top of the unit and the adjacent inside surfaces of the container (see Drawing 5-13-1172).

3.6.4.1.1 Base. The unit shall be bolted to the base of the shipping container with carriage bolts extending through the load-bearing members or through the skids and load bearing members, or through the skids of the base, and up through the base or legs of the unit. The heads of the carriage bolts shall be at the under-side of the floorboard, load-bearing members, or the skids. Not less than 4 bolts shall be used for securing the unit to the crate base. Bolt diameter shall be not less than 3/8-inch for units weighing up to 500 pounds. Units in excess of 500 pounds shall require 1/2-inch bolts. Alternative to carriage bolts, units weighing less than 400 pounds may be secured by minimum 5/16-inch diameter machine bolts. Round, flat washers and lockwashers shall be used under the heads of all machine bolts, and lockwashers shall be used with all nuts on carriage bolts.

3.6.4.1.2 Panel. The unit shall be secured to one panel of the container with two bolts, each bolt extending through a cleat on the outside face of the container, through the container panel, and through a back bar of group III or IV woods of MIL-STD-731, minimum 1-1/8 inches thick by 5-1/2 inches wide extending the full inside width of the container. Each bolt shall also extend through one of two blocking members of group III or IV woods, minimum 1-1/8 inches thick by 5-1/2 inches wide, blocking members extending to the inside surface of the top and side panels of the container, with bolts extending into gusset plates which form a part of the rear face of the unit. The extending of the bolts through a cleat on the outside surface of the container may be omitted provided that two bracing members, minimum 3/4-inch thick by 1-3/4 inches wide, are secured to the inside surface of the top or side panels of the container and abut the blocking members bolted to the rear face of the unit.

3.6.4.1.3 Edge pads. Edge pads shall be made of built up corrugated fiberboard. Refrigerators with top mounted units shall be provided with full length pads extending down from the top edge of the sides and front a minimum of 3 inches and of sufficient thickness to provide a tight pack and a minimum clearance of 1 inch between the outermost projections of the front and sides of the unit and the interior surface of the shipping container. All other units shall be provided with top edge and front and side edge pads. These pads shall be full length of the sides and front and shall extend in both directions (over the top and down the front and sides) a minimum of 3 inches. These pads shall be of the sufficient thickness to provide a tight pack and a minimum clearance of 1 inch between the top, front and sides of the unit and the interior surface of the shipping container. The pads shall be secured with tape conforming to PPP-T-97.

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3.6.4.2 Method B (pressure pack). The unit shall be cushioned and held securely within the shipping container by built-up, corrugated, fiberboard, edge pads. Edge pads shall be provided on all horizontal edges and on all vertical edges of the unit. Edge pads shall be of sufficient width to extend 3 inches in from the edge onto top, bottom, sides and ends of the unit, and of sufficient thickness to provide a tight pack and a minimum 1-inch clearance between any protruding part of the unit and the adjacent inside surfaces of the container.

3.6.4.3 Method C (blocking and bracing). A double-faced, corrugated, fiberboard tray, minimum 4 inches deep, shall be placed on the top and bottom of the refrigerator. The tray shall fit the top and bottom of the refrigerator in a compact manner. Blocking and bracing shall be in accordance with the appendix of MIL-C-104, and as specified herein. A perimeter wood frame of sufficient width and thickness to prevent side or end movement of the unit, shall be nailed or bolted to the base of the container. A perimeter frame shall also be secured to the underside of the top frame members to eliminate top sway. When possible, the unit shall be bolted to the base of the container with a minimum of four 1/2-inch bolts as specified in 3.6.4.1.1. A minimum of one inch clearance shall be provided between the sides and ends of the unit, or any protruding part in front of the unit, and the interior surfaces of the shipping container.

3.6.4.3.1 Alternative to method C. Each unit, with top mounted refrigeration units, shall be anchored in accordance with the appendix of MIL-C-104 or each unit with top mounted refrigeration units shall be bolted to the base of the crate with not less than four 1/2-inch diameter carriage bolts as specified in 3.6.4.1.1, and the base blocked with a perimeter wood frame of sufficient width and thickness to prevent side or end movement of the unit nailed to the base of the crate. Double-faced, corrugated, fiberboard, edge pads, having minimum width of three inches on each face, shall be placed between the bottom and finished surfaces of the sides and ends of the unit and wood frame and the base of the crate, and around the top edge of the cabinet extending down the sides and ends of the cabinet and be secured with tape conforming to PPP-T-97. The bottom fiberboard pads may be omitted when the perimeter wood frame contacts the structural steel base and not the finished surfaces of the sides and ends of the cabinet. A minimum clearance of one inch shall be maintained between any protruding part of the top, sides, and ends of the unit and the interior surfaces of the shipping container.

3.6.5 Protection of finished surfaces. One of the following shall be placed between the corrugated fiberboard or blocking members and the finished surfaces of the unit:

(a) A sheet of wax paper conforming to type III of UU-P-134.
(b) Cellulosic cushioning material conforming to type I or II, class B of PPP-C-843.

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The face of cushioning material in direct contact with fiberboard pads or blocking members shall be surfaced with kraft paper having a minimum basis weight of 30 pounds per ream (24 by 36-500).

3.7 Marking.

3.7.1 Interior packages. Each package, box, and overwrap shall be marked or labeled to indicate the item and quantity contained therein. Component parts that have been overwrapped and secured in their respective locations shall not require marking.

3.7.2 Shipping containers. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129. Set marking requirements shall apply when accessories and parts are packed separate from the unit.

3.7.2.1 Precautionary markings. Special handling marking requirements in MIL-STD-129, applicable to fragile items, arrows, the words "UP" and "THIS SIDE UP" and "CENTER OF BALANCE" shall apply.

3.7.2.2 Additional marking. The panel of the sheathed nailed wood crate, cleated fiberboard, cleated plywood, or nailed wood shipping container against the front of the refrigerator being packed shall be marked "FRONT" in letters not less than 1 inch in height.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check or test such items to determine the validity of the certification.

4.3 Inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions of MIL-STD-105.

4.3.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

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* 4.3.2 Examination of the end item. Examination of the end item shall be in accordance with the classification of defects set forth in tables II, III, IV, V, and VI, and the inspection levels and acceptable quality levels (AQL's) specified in 4.3.2.1. The sample unit for this examination shall be one packaged or packed refrigerator, or related equipment as applicable, of the specified type, design, and characteristic. The lot size shall be expressed in terms of the applicable sample unit for purposes of determining sample size. The inspection lot shall consist of all packaged or packed items submitted for inspection at one time.

TABLE II. Visual examination for preparation for packaging

Examine	Defect	Classification Major	Minor
Disassembly	Detachable parts and accessories not removed to prevent damage (when specified)	X	
	Detachable parts and accessories not secured in position (when specified)	X	
Matchmarking	Tags with information not attached to mating parts (when specified)		X
Draining	All water not removed from water containing parts of refrigerator	X	
Deprocessing tag	Unit not tagged to show deprocessing requirements		X

TABLE III. Examination of the end item for packaging

Examine	Defect	Classification Major	Minor
Level A			
Option I			
Sealing exposed ends of wire, plug, openings, sockets, faces of gages, and instruction plates	Not sealed with tape when specified		X

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TABLE III. Examination of the end item for packaging (cont'd)

Examine	Defect	Classification Major	Minor
Electrical components	Components not preserved as specified	X	
	Openings into interiors not taped		X
Operating instructions	Not placed individually in envelope		X
Refrigerating units	Drive belts not mounted in place with tension relaxed		X
Floating mechanism and tubing	Not secured in fixed position	X	
Doors, drawers and bins	Not secured as specified	X	
	Keys not bagged and secured to handle (when specified)	X	
	Cushioning material missing (when specified)	X	
	Strap not centered on cushioning material		X
	Strap tensioned too tightly causing damage	X	
	Strap tensioned too loosely causing slippage	X	
	Not taped as specified (when required)	X	
Legs	Not removed when detachable	X	
	Hardware missing from package	X	
	Not cushioned and packaged as specified	X	
Leveling pads	Not completely screwed in, or not detached and packaged, as specified		X
Line cord	Not looped and tied securely as specified		X
	Voltage and frequency notice missing or not attached (when specified)	X	
Accessories and parts (boxed)	Not individually wrapped with cushioning material	X	
	Not packaged in box specified	X	
	Box not closed as specified	X	

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TABLE IV. Examination of the end item for packing

Examine	Defect	Classification Major	Minor
Levels A and B			
Option I			
Refrigerator	Not secured and cushioned within container as specified (see table I and 3.6.4)	X	
	Finished surfaces not protected from abrasion with material specified	X	
Accessories and parts box	Not securely blocked and braced within refrigerator (when specified)	X	
	Not secured within the shipping container (when specified)	X	
	More than two shipping containers per complete refrigerator when packed separately	X	
Container	Not as specified for weight limits of refrigerators (see 3.6.1.1.1, 3.6.1.1.2, and 3.6.1.1.3)	X	
	Less than minimum clearance specified between refrigerator and inside of container	X	
Waterproof barrier	Waterproof barrier missing (when specified)	X	
	Less than 4 inch lap at joints		X
Bolts and nuts	Less than four bolts holding refrigerator	X	
	Diameter of bolt less than specified	X	
	One or more flat washers missing under heads of machine bolts	X	
	One or more nuts loose on bolts	X	
	One or more lockwashers missing	X	
	Bolts not positioned through container members and blocking as specified	X	

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TABLE IV. Examination of the end item for packing (cont'd)

Examine	Defect	Classification Major	Minor
Blocking members	Dimensions less than specified	X	
Edge pad	Not built-up corrugated fiberboard	X	
	Less than length required to cover edge	X	
	Less than three inches of pad contacting each adjacent corner surface of refrigerator	X	
	Does not provide minimum one inch clearance beyond outermost projection	X	
	Not fitted as required to make tight pack	X	
Tray	Not double faced corrugated	X	
	Less than 4 inches deep	X	
	Missing on top or bottom	X	
Wood frame	Width and thickness less than specified	X	
	Not secured to base as specified	X	
	Missing on top or bottom	X	

TABLE V. Examination of the end item for marking

Examine	Defect	Classification Major	Minor
Marking			
Package and shipping container, as applicable	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application	X	

4.3.2.1 Inspection levels and AQL's for examination. The inspection levels, for determining the sample size, and the AQL's expressed in defects per hundred units shall be as follows:

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Table No.	Inspection levels	AQL's Major	AQL's Total
II	S-2	2.5	10
III	S-2	2.5	10
IV	S-2	2.5	10
V	S-2		6.5
VI	S-2		4.0

4.3.3 Examination of shipping container. When shipping containers are required to be in accordance with PPP-B-621, PPP-B-636, PPP-B-640, PPP-C-650, MIL-C-104, and MIL-C-11133, examination for closure and strapping shall be in accordance with the applicable specification.

4.3.3.1 When shipping containers are required to be in accordance with PPP-B-591 and PPP-B-601, examination of closure and strapping or determination of compliance with the specification or appendix of the specification, as applicable, and Drawing 5-13-1153 of this specification shall be in accordance with the defects in table VI.

TABLE VI. Examination for strapping and closure of the shipping container conforming to PPP-B-591 and PPP-B-601

Examine	Defect	Classification Major	Classification Minor
Strapping	Not applied as specified (Drawing 5-13-1153)	X	
	Missing strap	X	
	Loose strap	X	
	Torn strap	X	
	Cut strap	X	
	Not drawn tight	X	
	Weak joint	X	
Staples	Staples less than 4 inches from edge of member		X
	Less than size specified	X	
	Not spaced as specified		X

4.3.4 Testing of the end item. The end item shall be tested for compliance with 3.5.1.1.1, 3.5.1.1.3, 3.5.1.1.5, and 3.5.1.1.8, as specified in 4.4. The sample unit shall be one unit. The lot for determining the sample size shall be expressed in terms of sample units. The inspection level shall be S-2 and the AQL shall be 6.5 expressed as percent defective. Test failures of a determination shall indicate a defective unit.

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4.3.4.1 Certificate. When a supplier of components has tested materials in accordance with the applicable referenced material specification, a certificate of compliance from the packaging material supplier to the commodity supplier will satisfy the physical requirements (see 4.2).

4.4 Tests.

4.4.1 Staining test. The staining test shall be performed as specified in 4.4.1.1, 4.4.1.2, and 4.4.1.3.

4.4.1.1 Conditioning. Tape shall be conditioned for testing for not less than 24 hours at 73 degrees Fahrenheit (F), plus or minus 2F, and 50 percent plus or minus 4 percent, relative humidity.

4.4.1.2 Specimens. The first three layers of tape shall be removed from the rolls before taking specimens to be tested. An equal number of 10 inch lengths of tape shall be taken as specimens from all sample rolls to be tested.

4.4.1.3 Staining. Specimens of tape shall be applied, adhesive down, to a clean porcelain enamel surface panel and rolled down with two passes, once in each longitudinal direction, using a 4-1/2 pound rubber roller, having a surface durometer hardness between 70 and 80, and approximately 3-1/2 inches in diameter, at a rate of travel of 10 inches per minute. Panels shall be placed in an oven having a temperature of 150F, plus or minus 2F, for a period of 10 days. Upon removal, panels shall be allowed to cool 2 hours before stripping off the tape. Tape edge transfer may be noticed due to the high temperature. If so, it should be removed with solvent prior to determination for staining. Tape shall be rejected for use if staining is evident.

4.4.2 Cleaning, preservation, and unit process protection. Tests of cleaning, preservation, and unit protection processes shall be in accordance with the requirements of MIL-P-116.

5. PREPARATION FOR DELIVERY

5.1 Not applicable.

6. NOTES

6.1 Intended use. The preservation, packaging, packing and marking requirements of this specification are intended for the preparation for delivery of refrigerators and related equipment in accordance with either of two options (see 3.5.1, 3.6.1 and 3.6.2).

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation, packaging, and packing (see 3.5 and 3.6).
(c) When weather resistant fiberboard is required (see 3.6.2.1.1, 3.6.2.1.2, and 3.6.2.1.3).

6.3 Level C, packing. Level C packing should be specified only for direct domestic shipment from the supply source to the first receiving activity for immediate use.

6.4 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - YD
Air Force - 82

Preparing activity:

Army - GL

Project No. 4110-0233

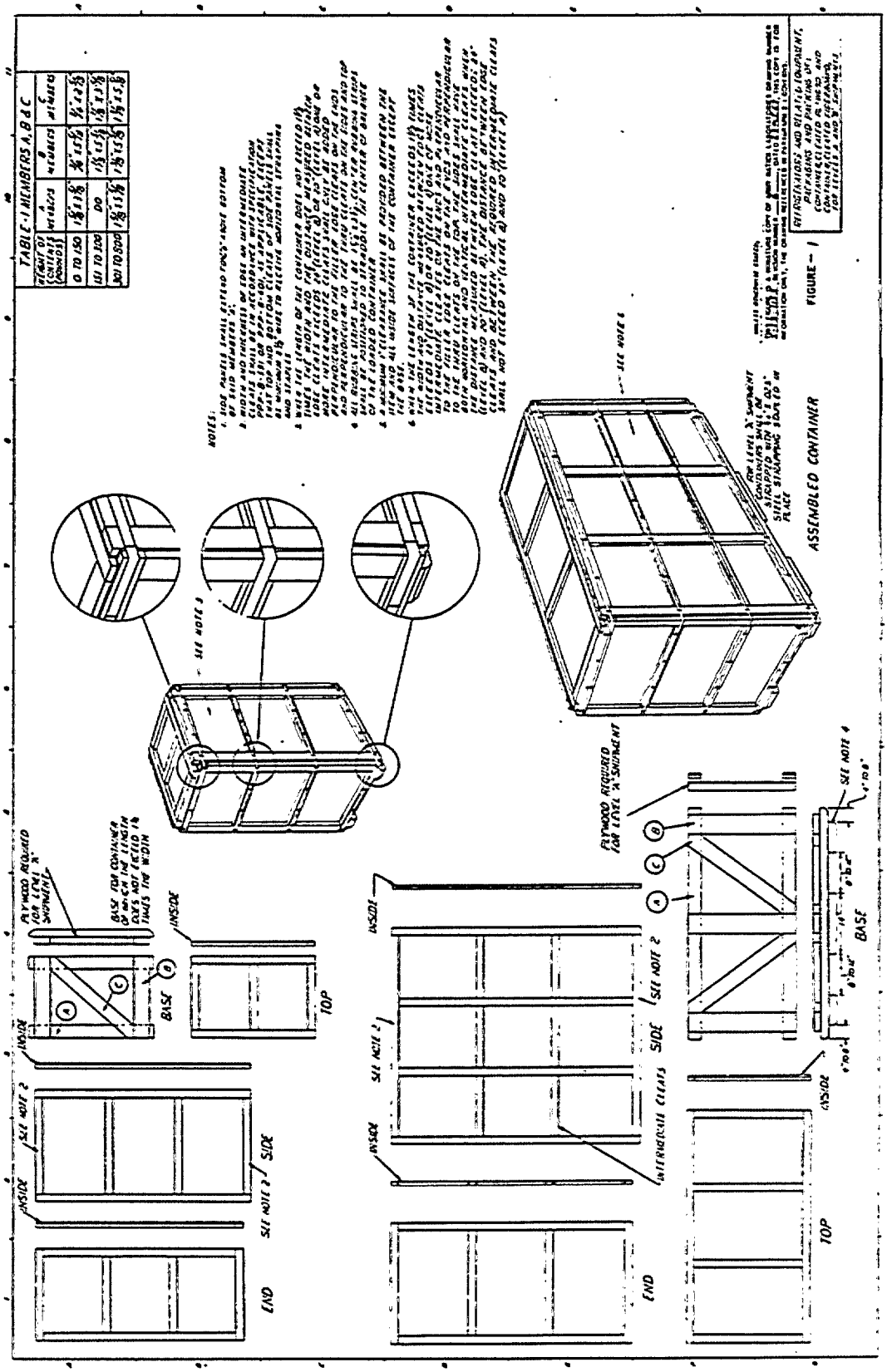
Review activities:

Army - EA
DSA - DGSC-SEB

User activities:

Army - ME
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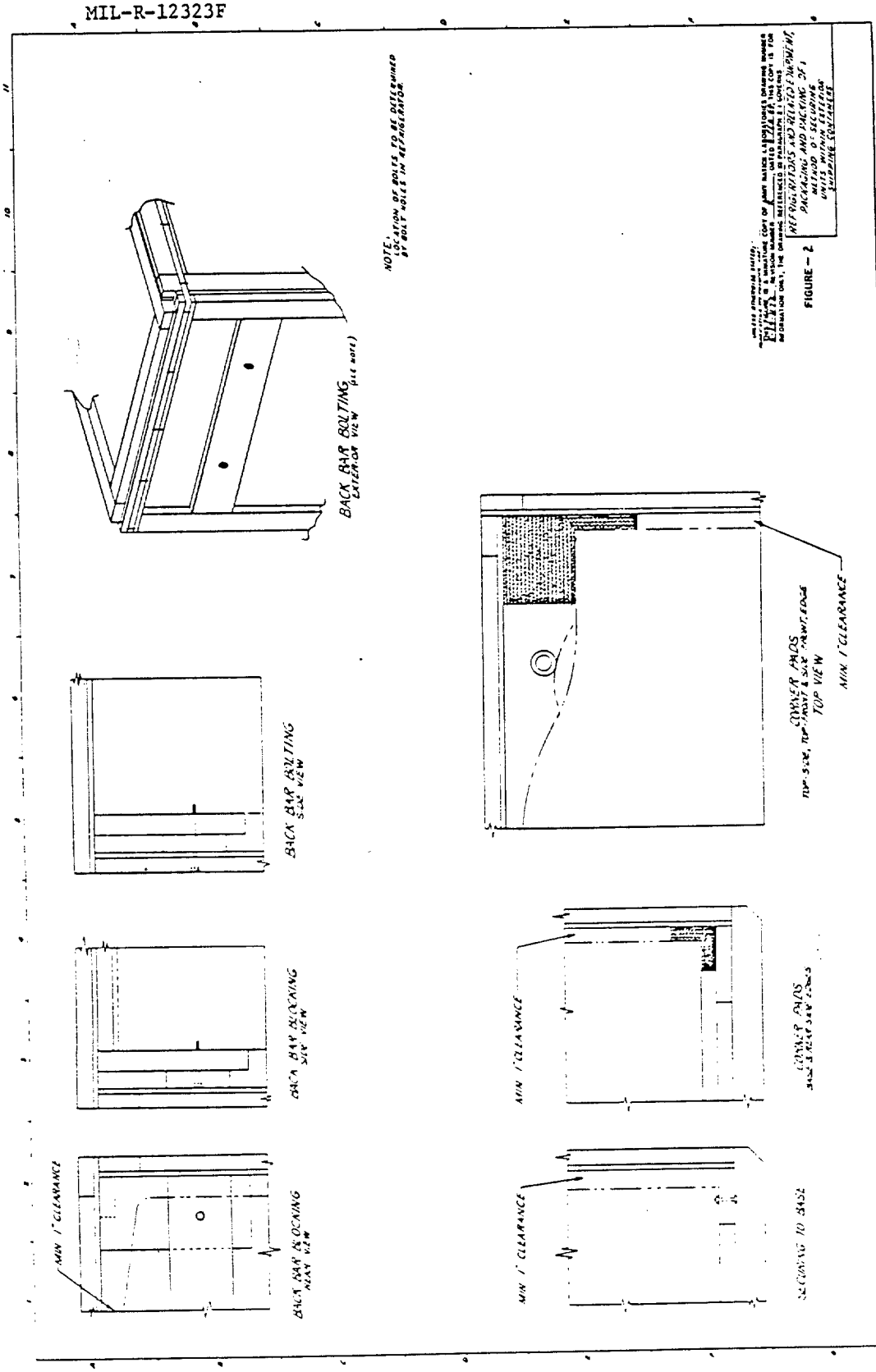


FIGURE – 2

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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